## TIER 1 — edge controller (per floor, per room, every 5 s)

readings window
smoke · CO · temperature (deque per sensor)

sensor stale?
value unchanged ≥ 60 s

smoke ≥ 10
and CO ≥ 20 ?

yes

no

CO stale and
smoke ≥ 20 ?

yes

no

FIRE — release every door on this floor
(first fire only), alert server

smoke ≥ 5 ?

yes

isolation forest (ONNX)
anomaly ? p50 18 ms

anomaly

no

smoke ≥ 10 ?

yes (safety net)

no

POSSIBLE — alert server
(no local action)

normal

publish clear when a candidate
drops below threshold

no

## TIER 2 — server agent (per alert, 3 in parallel)

dedupe
same room + severity within 60 s → drop
confirmed room → never re-assessed

features: readings, rates, faulty-sensor list,
spread risk (door graph), evacuation route

LLM verdict {is_fire, confidence, reasoning, alert}
Qwen2.5-14B · p50 2.1 s
(unreachable → rule-based fallback)

is_fire and
confidence ≥ 0.7 ?

yes

CONFIRM
release doors on the other floors (once per incident)
evacuation routes · viewer effects + red card
log confirm

no

WATCH 60 s
orange card · log watch_pending

after 60 s, live sensors:
smoke < 2 and CO < 5 ?

no — keep watching

no

smoke ≥ 10 and CO ≥ 20 ?

yes

ESCALATE on readings
(same as confirm)

yes

confirmed room:
readings clear for 60 s ?

yes

STAND-DOWN to the edge
floor_clear · relock
remove card

when no room is
confirmed or watched

ALL CLEAR
lock every floor · clear viewer

alerts/<floor>/<room> severity=confirmed|critical

alerts/<floor>/<room> severity=warning

commands/<floor>/stand-down
(edge re-locks if floor clear and it owns the release)